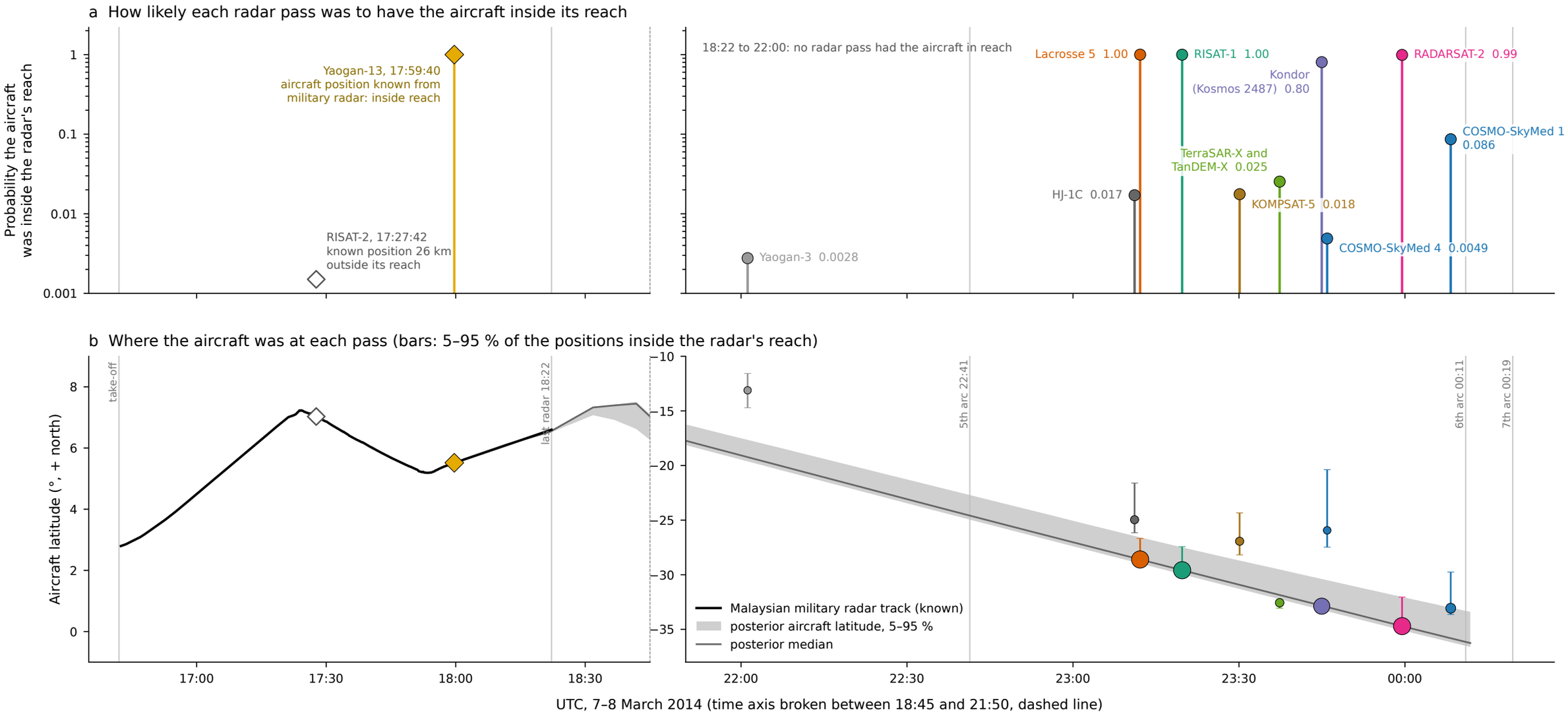
# Radar satellites that could have imaged MH370 in flight, 7–8 March 2014

A radar pass counts when the line straight across the satellite's track swept over the aircraft's position while that position lay inside the radar's range of incidence angles, on a side it can look to.




Plain-language note: four radar satellites passed while the aircraft was almost certainly inside the area they could image: Lacrosse 5 (USA) at 23:12, RISAT-1 (India) at 23:20, Kondor, Kosmos 2487 (Russia) at 23:45, and RADARSAT-2 (Canada) at 23:59 UTC. No radar had the aircraft in reach between 18:22 and 22:00. At 17:59 the aircraft's position is known from the Malaysian military radar, and it lay inside the reach of the Chinese radar Yaogan-13; that is the only pass where the aircraft's position is certain, so it is the test case for whether such an image would show an airliner. This shows what could have been imaged, not what was imaged: a radar records only when ordered to, and each image covers one strip inside its reach.

Technical: posterior positions from core run J routes.npy (2,000 posterior routes per seed, 10-min vertices 18:01:49–00:11:49, linear interpolation; waypoints family 48,000 routes, P(family) 0.9982; probabilities are the four-family mixture). Known positions: Geoscience Australia digital radar track 16:42:27–18:22:12 (indicative times). SGP4 on March 2014 element sets, 5 s steps; zero-Doppler crossing per route by sign change of the along-track coordinate. Incidence ranges: RADARSAT-2 10–60°, RISAT-1 12–55°, RISAT-2 20–45°, COSMO-SkyMed 20–60° (grade B); Lacrosse 5 20–60°, Kosmos 2487 20–55°, Yaogan 20–55° (open-literature estimates, grade C). Lacrosse 5 and the KH-11s use amateur element sets (epochs 27 Feb and 18–19 Mar 2014). Passes with probability below 0.001 are not drawn.